



Southeast University

Department of Computer Science and Engineering

LAB Report

Course Name: Computer Networking LAB

Course Code: CSE342.2

Experiment No.: 07

Experiment Name: Dynamic Routing - RIPV2 Configuration and Protocol Analysis.

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❑ Experiment No: 07

❑ Experiment Name: Dynamic Routing - RIPv2 Configuration and Protocol Analysis.

❑ Introduction: This lab experiment explores the configuration and verification of the Routing Information Protocol version 2 (RIPv2) within a simulated environment. Dynamic Routing Protocols are essential for enabling communication across multiple segmented networks without the need for manual static route administration. RIPv2 is deployed across a core network of three series routers to facilitate automated route discovery adapt to topology changes and established full end-to-end connectivity between four distinct class C subnets.

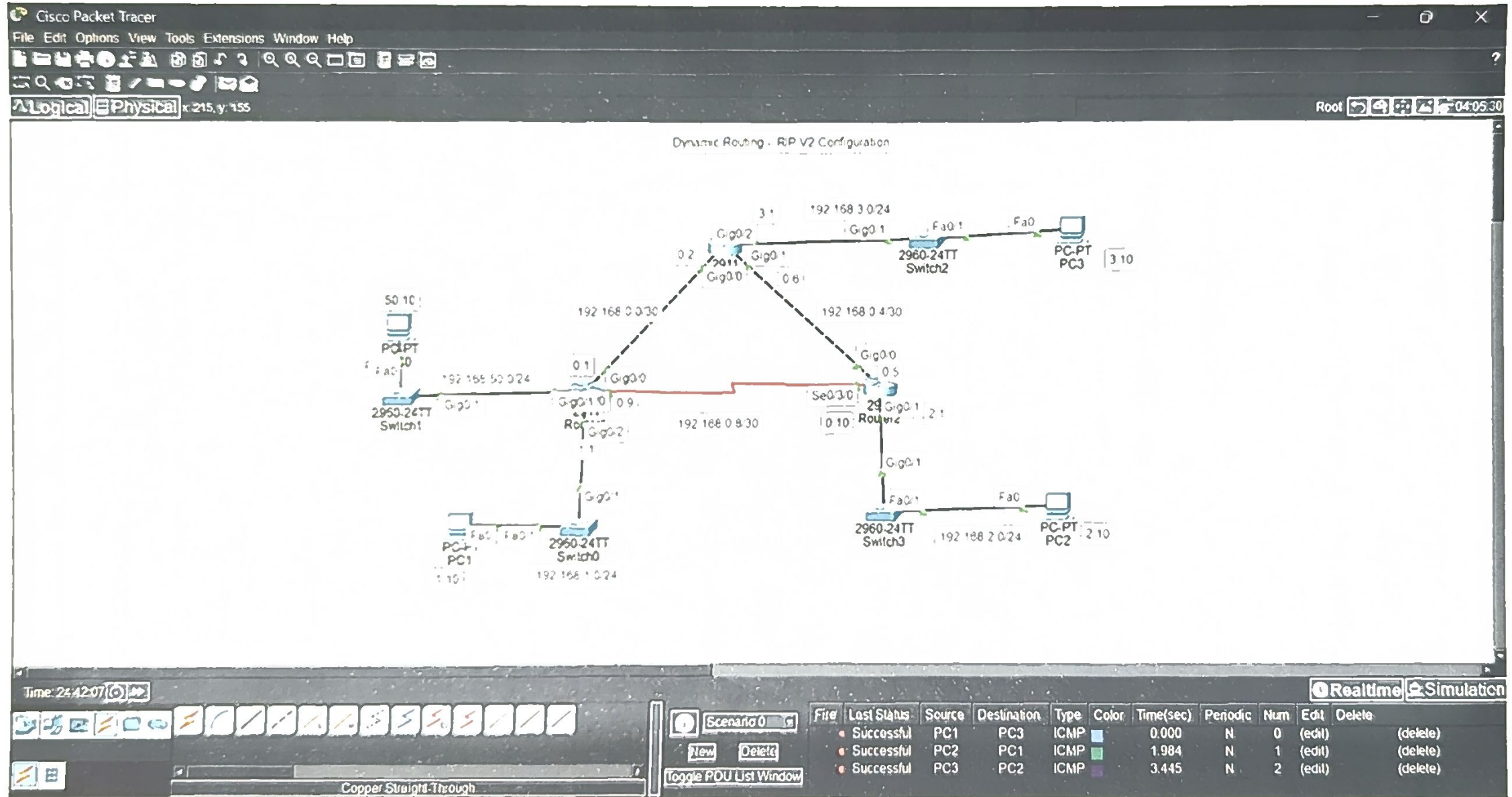
❑ Objectives:

- To Implement and verify Dynamic Routing using Routing Information Protocol (RIPv2) across a multi router topology.
- To troubleshoot and resolve Layer 1 / Layer 2 Connectivity Issues, including administratively down interfaces and DCE Clock rate configuration.

- To critically analyze RIPv2 behavior focusing on three key Parameters: Information sharing mechanisms, Path Calculation and network Convergence time.
- To establish end-to-end ICMP reachability between logically and Physically separated internal subnets.

Network Addressing Table :

Device	Interface	IP Address / subnet	Default Gateway
Router1 (Left)	Gig 0/1	192.168.1.1/24	N/A
Router1 (Left)	Gig 0/0	192.168.50.1/24	N/A
Router1 (Left)	Gig 0/2	192.168.0.1/30	N/A
Router1 (Left)	Se 0/3/0 (DCE)	192.168.0.9/30	N/A
Router0 (Top)	Gig 0/0	192.168.0.2/30	N/A
Router0 (Top)	Gig 0/2	192.168.3.1/24	N/A
Router0 (Top)	Gig 0/1	192.168.0.6/30	N/A
Router2 (Right)	Gig 0/1	192.168.2.1/24	N/A
Router2 (Right)	Gig 0/0	192.168.0.5/30	N/A
Router2 (Right)	Se 0/3/0 (DCE)	192.168.0.10/30	N/A
PC1	Fa 0/0	192.168.1.10/24	192.168.1.1
PC0	Fa 0/0	192.168.50.10/24	192.168.50.1
PC2	Fa 0/0	192.168.2.10/24	192.168.2.1
PC3	Fa 0/0	192.168.3.10/24	192.168.3.1



☐ Device Configuration Specifications: The following configuration Address the offline Interface states and fully implement the RIPv2 dynamic Routing Protocol.

- Router 1 (Left Router):

Router > Enable

Router # Configuration terminal

Router(Config)# Hostname Router1

Router1(Config)# Interface gig 0/2

Router1(Config)# No shutdown

Router1(Config)# Exit

Router1(Config)# Interface Serial 0/3/0

Router1(Config)# Clock rate 64000

Router1(Config)# no shutdown

Router1(Config)# Exit

Router1(Config)# router rip

Router1(Config)# version 2

Router1(Config)# Network 192.168.1.0

Router1(Config)# Network 192.168.50.0

Router1(Config)# Network 192.168.0.0

Router1(Config)# Network 192.168.0.8

Router1(Config)# End

- Router 0 (Top Router):

Router > Enable

Router# Configure Terminal

Router(config)# Hostname Router 0

Router0(config)# router rip

Router0(config)# version 2

Router0(config)# Network 192.168.3.0

Router0(config)# Network 192.168.0.0

Router0(config)# Network 192.168.0.4

Router0(config)# End

- Router 2 (Right Router):

Router > Enable

Router# Configure Terminal

Router(config)# Interface serial 0/3/0

Router(config)# no shutdown

Router(config)# Exit

Router2(config)# router rip

Router2(config)# version 2

Router2(config)# Network 192.168.2.0

Router2(config)# Network 192.168.0.4

Router2(config)# Network 192.168.0.8

Router2(config)# End

□ Protocol Analysis: As Per the RIPv2 lab requirement the dynamic Routing Protocol was evaluated on three distinct Operational Parameters within the topology.

- Information Sharing Mechanisms: RIPv2 is classified as a distance-vector routing Protocol. It disseminates its network knowledge by Periodically transmitting its entire Routing table out of all RIPv2-enabled Interfaces. Unlike its Predecessor (RIPv1) which relies on global network broadcasts, RIPv2 utilizes the multicast IP Address 224.0.0.9. While multicast is more efficient for non-RIP host devices, the Periodic nature of these updates (every 30 seconds) inherently consumes consistent link ~~band~~ bandwidth. which scales Poorly in expansive, stable networks compared to the triggered-update mechanisms of link-state Protocols.

- Path Calculation and metric selection: RIPv2 Calculates Optimal network Paths utilizing a one-dimensional metric. The Algorithm calculates the direct Point-to-Point Serial link as the Primary Path (1 hop distance). RIPv2 inherently fails to account for link bandwidth, delay or reliability resulting in traffic being forwarded over a slower Serial Interface simply because it represents fewer Physical routing hops.

• Network Convergence Time: Convergence dictates how swiftly a routing domain synchronizes its routing table following a topology alteration or link failure. RIPv2 demonstrates extremely sluggish convergence governed by standardized timers. An invalid time 180 seconds and a flush timer 240 seconds. Should the Primary serial link fail, the routers will not immediately utilize the alternative Gigabit Path. ~~They~~ Instead, they will wait up to three minutes to declare the route dead before flushing the invalid route and establishing the secondary Path. This delay causes ~~severe~~ severe packet loss during failover scenarios.

☐ Verification Protocols: To certify standard Operational Parameters and Inter-Subnet Connectivity, the following Validation Procedures must be executed.

Router# show ip route database

Router# show ip route rip

Execute these commands on Routers to verify RIPv2 Routing table.